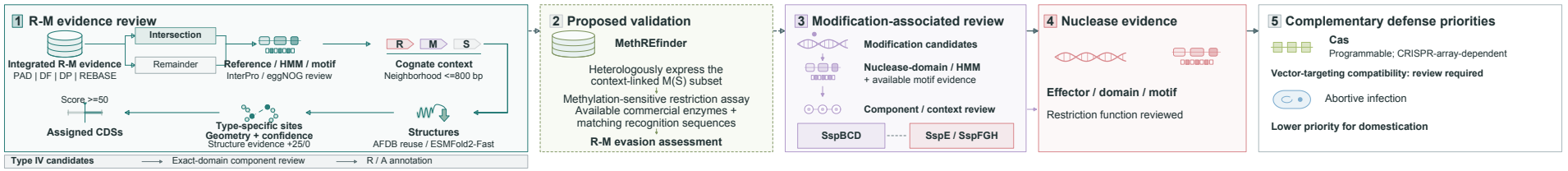
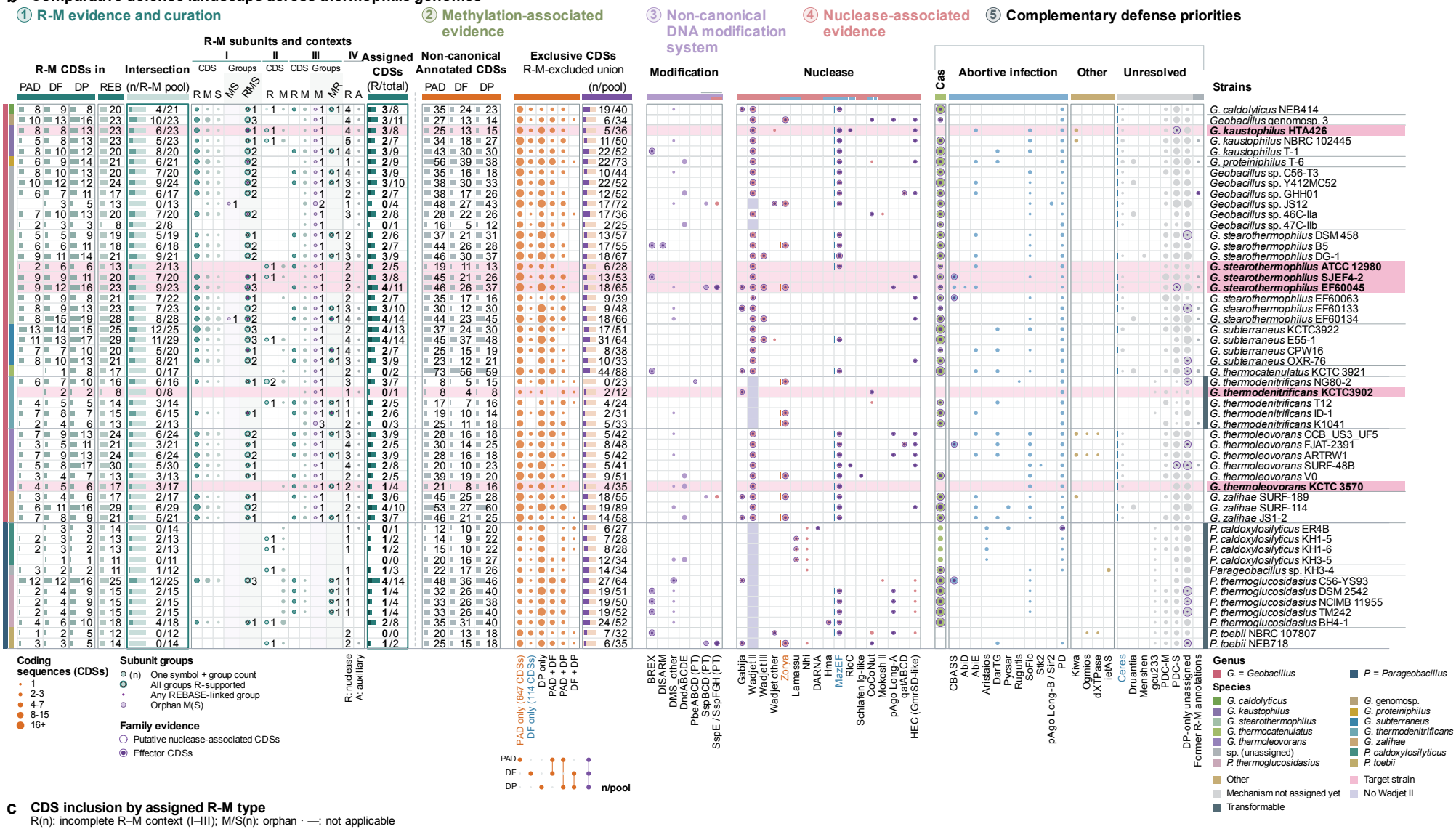
a Evidence criteria for DNMB-guided barrier prioritization



b Comparative defense landscape across thermophilic genomes



c CDS inclusion by assigned R-M type

R(n): incomplete R–M context (I–III); M/S(n): orphan · —: not applicable

G. stearothermophilus **EF60045**

Type	Pool	Shared-pool CDSs: PAD	DF	DP	DNMB components: DNMB	R	M	S	A
I	9	9	9	9	9	3	3	3	0
II	0	0	0	0	0	0	0	—	0
III	3	2	2	2	2	1 (1)	1 (1)	—	0
IV	3	1	1	3	3	2	0	0	1
Total	**15**	**12**	**12**	**14**	**14**	**6 (1)**	**4 (1)**	**3**	**1**

G. stearothermophilus **SJEF4-2**

Type	Pool	Shared-pool CDSs: PAD	DF	DP	DNMB components: DNMB	R	M	S	A
I	7	7	7	6	5	2 (1)	2	1	0
II	2	2	2	1	2	1	1	—	0
III	1	0	0	0	1	0	1 (1)	—	0
IV	2	1	0	1	2	2	0	0	0
Total	**12**	**10**	**9**	**8**	**10**	**5 (1)**	**4 (1)**	**1**	**0**

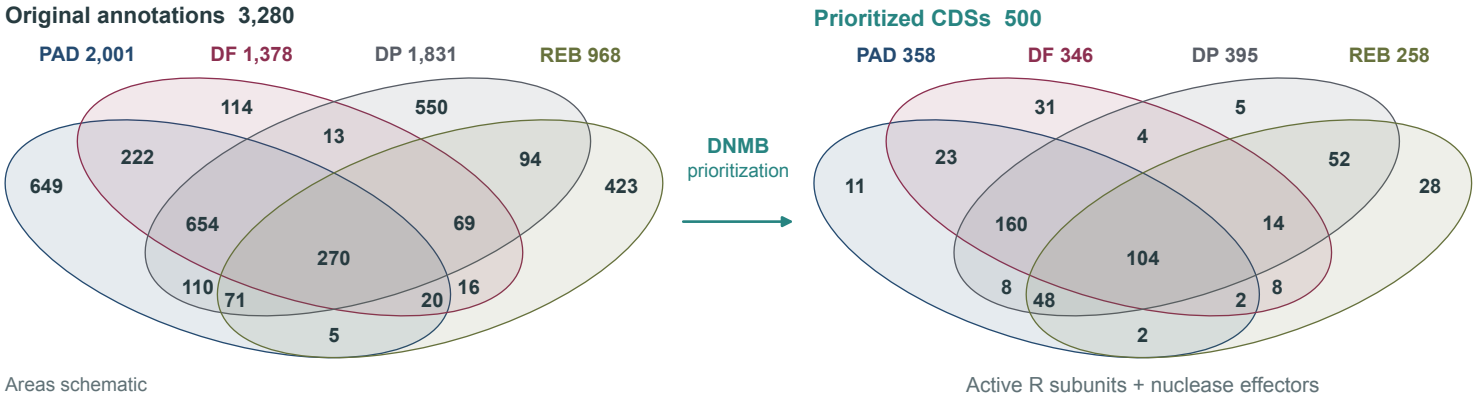
B. coagulans **DSM 1 (external)**

Type	Pool	Shared-pool CDSs: PAD	DF	DP	DNMB components: DNMB	R	M	S	A
I	6	6	6	6	6	2	2	2	0
II	1	0	0	0	1	0	1 (1)	—	0
III	2	2	2	2	2	1	1	—	0
IV	3	1	1	3	3	2	0	0	1
Total	**12**	**9**	**9**	**11**	**12**	**5**	**4 (1)**	**2**	**1**

Defense annotation comparison

52 genomes · Coding sequences (CDSs)

a DNMB prioritization



Areas schematic

Active R subunits + nuclease effectors

b Defense categories

Category	Number of genomes	DNMB	Putative nuclease-associated CDSs Total (effector CDSs)		DF	PAD	DP
R-M + IV-related	52	540	422	(232)	365	333	425
Modification	31	127	63	(10)	61	127	82
Nuclease (family pool)	51	352	329	(132)	287	283	227
Cas	42	501	481	(104)	467	500	394
Abortive infection	50	213	23	(6)	118	204	36
Other mechanisms	7	15	0	(0)	13	15	0
Mechanism unassigned	52	1,192	15	(15)	73	558	662
Former R-M residual	11	12	1	(1)	0	4	11

c Defense families

Family / pool	Number of genomes	DNMB	Putative nuclease-associated CDSs Total (effector CDSs)		DF	PAD	DP
R-M + IV-related	52	540	422	(232)	365	333	425
BREX	8	48	48	(8)	37	48	45
DISARM	1	5	5	(1)	5	5	5
DMS_other	18	26	3	(1)	2	26	11
DndABCDE	7	37	0	(0)	7	37	14
PbeABCD (PT)	1	1	1	(0)	0	1	1
SspBCD (PT)	4	12	6	(0)	12	12	8
SspE / SspFGH (PT)	4	10	6	(6)	10	10	10
Gabija	9	20	20	(10)	20	20	17
Wadjet II	33	132	132	(33)	131	132	130
Wadjet III	4	16	16	(4)	16	16	12
Wadjet other	4	6	4	(2)	0	6	2
Zorya	11	45	45	(11)	18	45	4
Lamassu	3	9	9	(3)	9	9	9
AbiO-Nhi	6	6	0	(0)	0	6	6
DARNA	1	1	1	(1)	1	1	1
Hma	1	3	3	(1)	3	3	3
MazEF	29	60	60	(30)	60	0	0
RloC	2	2	2	(2)	2	2	2
Schlafen Ig-like	1	1	1	(1)	1	0	0
CoCoNut	7	8	4	(4)	8	0	7
Mokosh II	2	2	0	(0)	0	2	2
pAgo Long-A	8	8	8	(8)	8	8	3
qatABCD	2	4	4	(2)	0	4	0
HEC	19	19	14	(14)	0	19	19
CRISPR-Cas	42	501	481	(104)	467	500	394
CBASS	5	24	22	(5)	24	24	15
AbiD	1	1	0	(0)	1	1	1
AbiE	16	32	0	(0)	32	32	0
Aristaios	4	6	0	(0)	6	3	6
DarTG	11	22	0	(0)	22	20	9
Pycsar	2	4	0	(0)	2	2	1
Rugutis	1	1	0	(0)	1	0	1
SoFic	31	43	0	(0)	26	43	0
Stk2	1	1	0	(0)	1	0	0
pAgo Long-B / Sir2	1	2	0	(0)	2	2	2
PD	46	77	1	(1)	1	77	1
Kiwa	5	10	0	(0)	10	10	0
Ogmios	3	3	0	(0)	3	3	0
dXTPase	3	3	0	(0)	3	3	0
ietAS	1	2	0	(0)	0	2	0
Ceres	28	33	0	(0)	33	0	0
Druantia	6	26	0	(0)	20	26	21
Menshen	1	2	0	(0)	0	2	2
gcu233	15	15	0	(0)	15	0	0
PDC-M	40	122	0	(0)	0	122	33
PDC-S	52	408	3	(3)	5	408	20
Unassigned (DP-only)	52	586	12	(12)	0	0	586
Former R-M residual	11	12	1	(1)	0	4	11

Putative nuclease-associated CDSs Effector CDSs (497) Rows overlap

Track: row union · Activity untested · PAD: PADLOC; DF: DefenseFinder; DP: DefensePredictor; REB: REBASE